Artificial Intelligence Institute for Cosmic Origins
Funded by the U.S. National Science Foundation and the Simons Foundation

cosmicAI Workshop Day 1

Data prep and random forests

Goals for today

1. Start a project – make progress – every incremental bit of progress *is progress*
2. You will work with a group and you will all help each other out
3. Ask questions no matter how small how big
4. Make sure you understand the basic motivation as you start to look at your data etc

Day 3
Intro Team Activity (problem, resources, form teams, and help groups start work)
Break
Team Work Time
LUNCH
Team Work Time
Group Presentations & Wrap Up

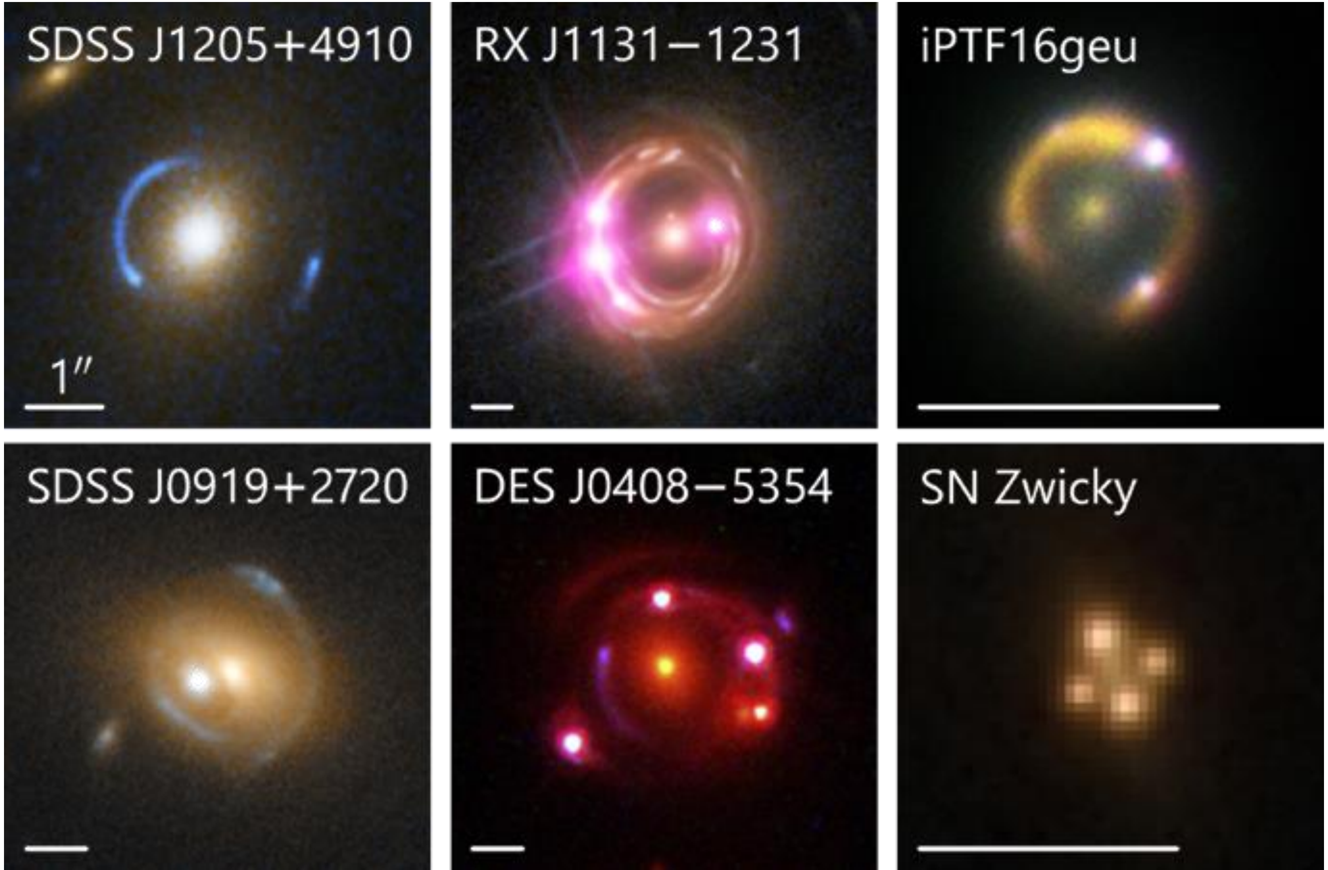
Coding axioms

1. **Making new cells should be your new best friend**
2. No matter what you are working with, you should look at a variable you have just defined (whether its a list, or an array, or a pandas dataframe) – make new cells and *look at what you're working with*
3. There is no one right way to do things – if the skeleton doesn't work for you, open new copy of the notebook and remove the skeleton!

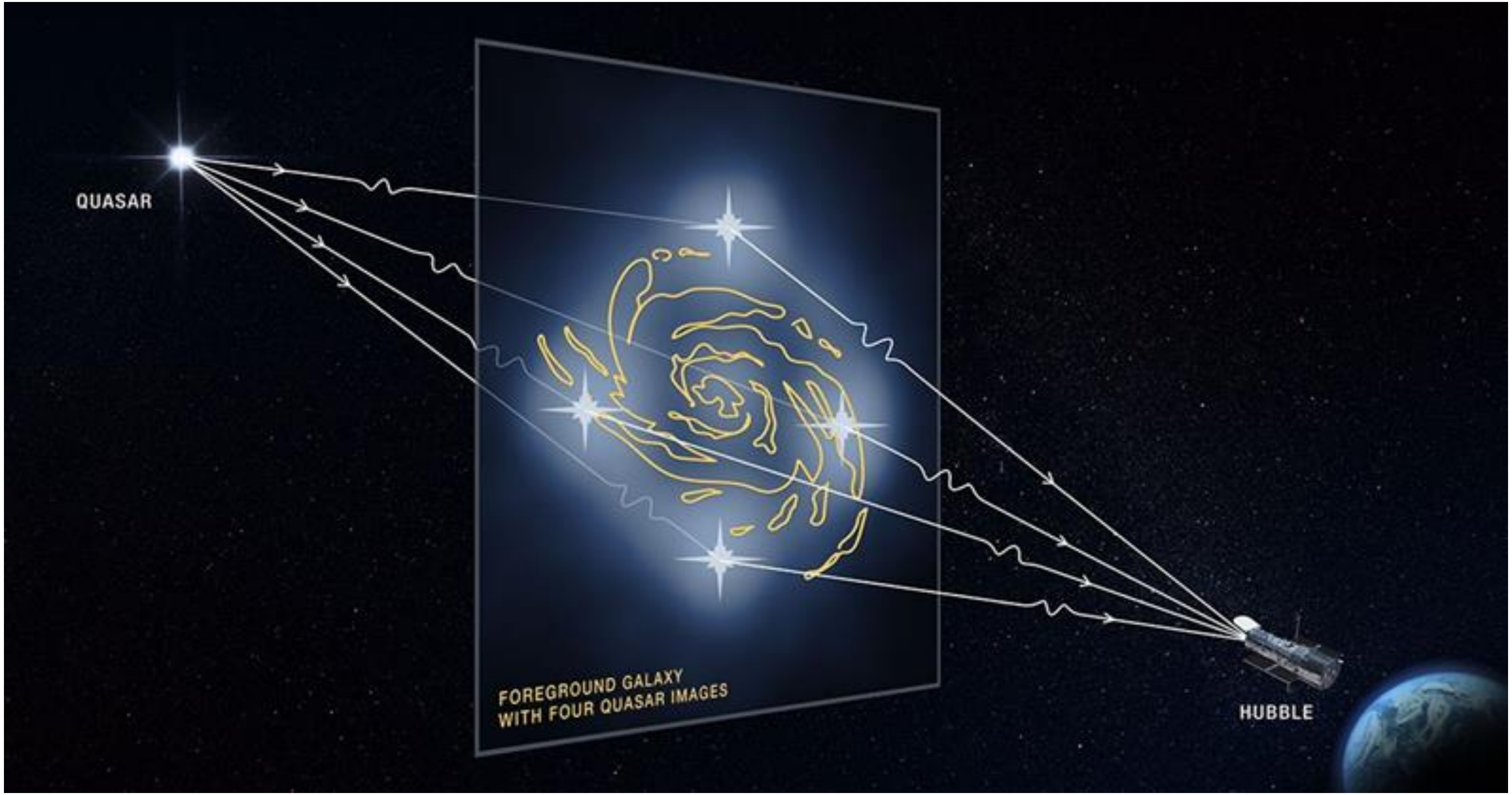
Projects overview

1. In classifying_alerts we look at how to find strongly lensed active galactic nuclei (AGN) from the Vera Rubin Observatory alert stream.
   a. These systems help us study cosmology, AGN structure, strong lens galaxies!
2. In tde_host_galaxy we try to find galaxies that are most likely to host black hole that tidally disrupt stars! There are some galaxies where we see tidal disruption events (TDE) more often (such as post-starburst galaxies).
   a. A TDE is when a star is broken down by a black hole at the center of a galaxy -- the stellar debris can form a disk around the star, affect star formation in the galaxy, or it can preferentially happen in galaxies that have had a lot of star formation happen recently -- anything could be true!

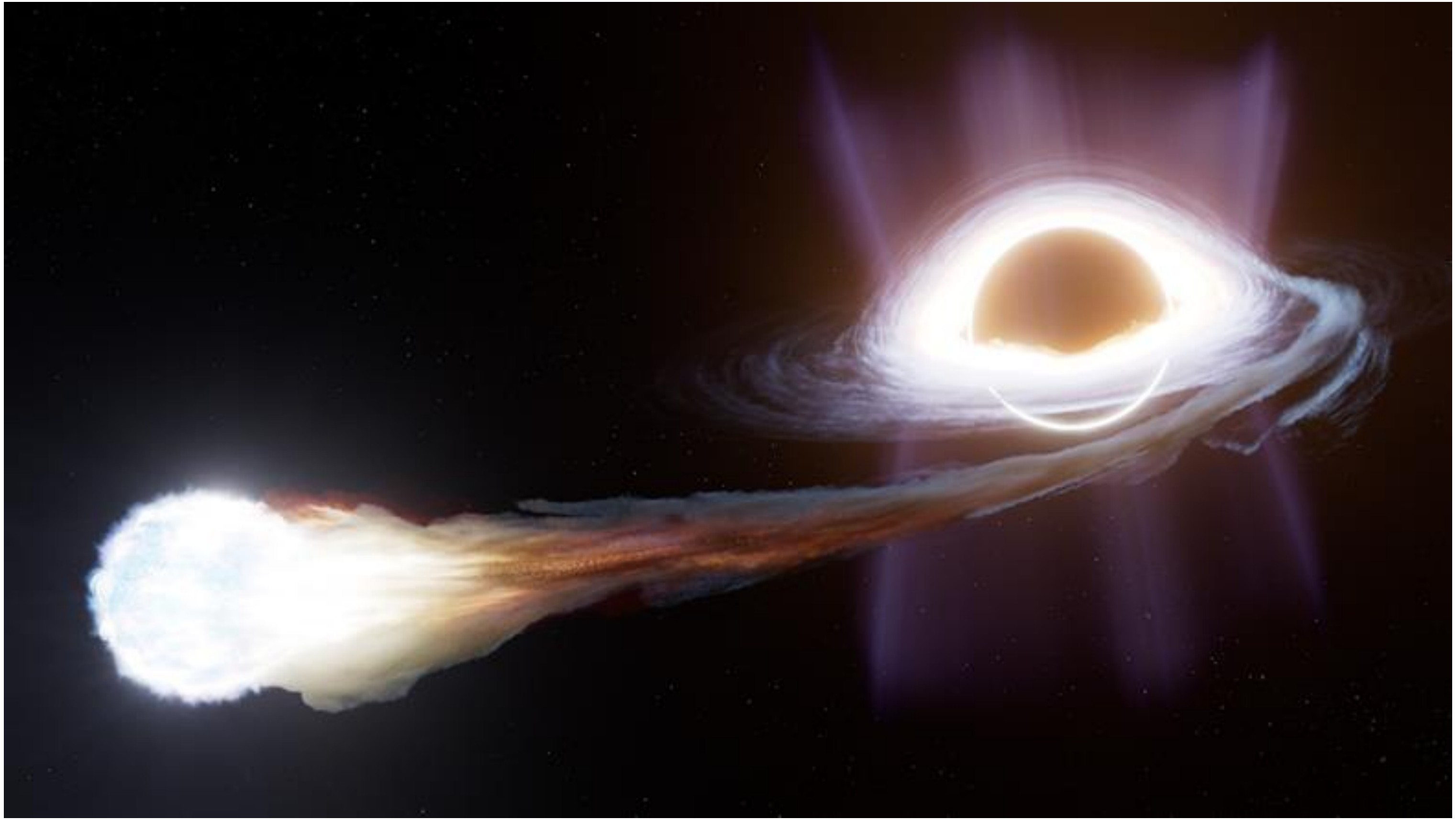
Strongly lensed AGN



Strongly lensed AGN



Tidal disruption events



Useful links for classifying alerts:

1. Extendedness features: https://github.com/drphilmarshall/lantern/blob/main/notebooks/1_dp1_diasource_characterization.ipynb
2. DIASource columns: https://sdm-schemas.lsst.io/dp1.html#DiaSource
3. https://docs.google.com/presentation/d/1cweC0QvG8tYI1bo9FECRe6Q0PnF4WnU5uSscy58Xrkc/preview

Useful links for TDE Host galaxies

1. French et al: https://iopscience.iop.org/article/10.3847/1538-4357/aaea64/pdf
2. Moon et al: https://iopscience.iop.org/article/10.3847/2515-5172/ad9822

Useful figure for TDE Host galaxies:

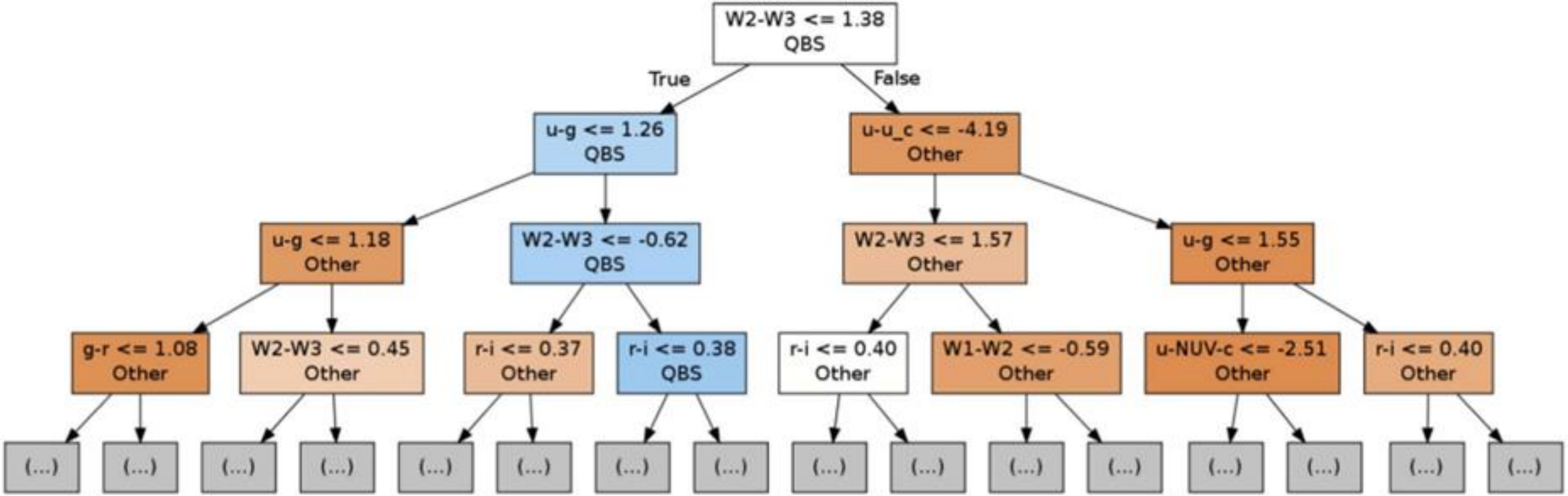
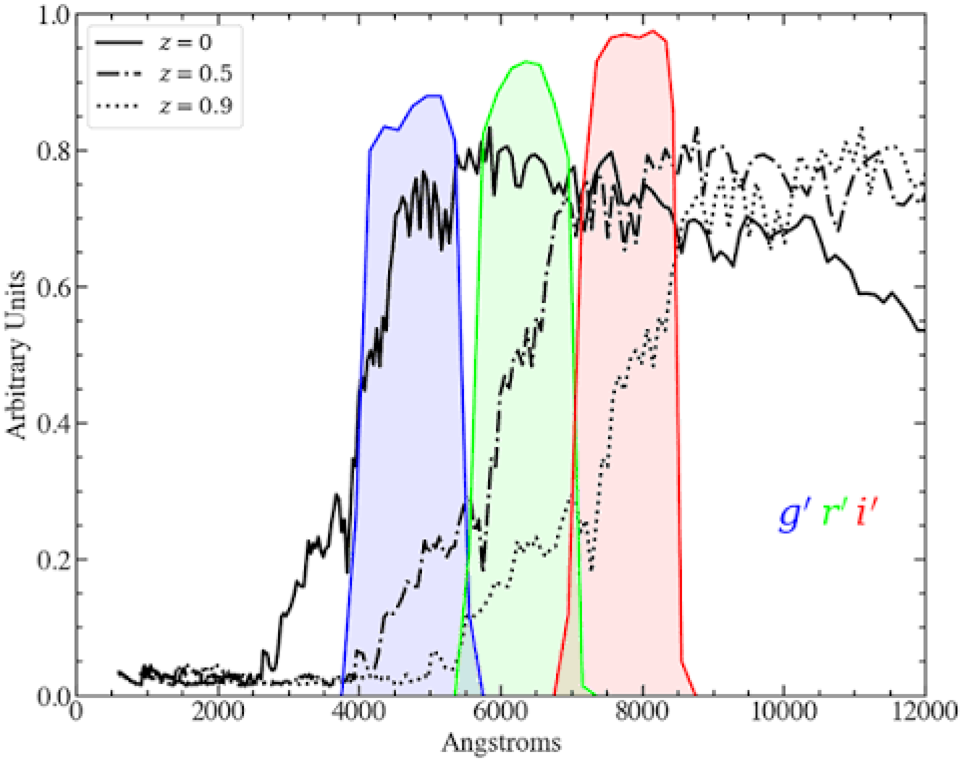


Figure 4. Example decision tree from the random forest for classifying quiescent Balmer-strong galaxies from imaging data alone. To generate each tree, galaxy observable features like color or concentration are chosen randomly, and the cut that maximizes the Gini impurity is chosen at each node. The example tree is truncated after three levels, but each tree in the random forest is grown until all leaves are pure, consisting of only one class. The random forest can then be used to classify galaxies, with each decision tree casting a vote on the galaxy classification. In this example, cuts are shown to distinguish quiescent Balmer-strong hosts from other galaxies. Blue nodes contain a higher fraction of quiescent Balmer-strong galaxies, and orange nodes contain a higher fraction of non-quiescent Balmer-strong galaxies.

TDE spectra → photometry



Group presentations

Project title + group name

1. What question were you trying to answer today?
2. What did you learn about the project?
    a. Show us plots!
    b. A good slide will include an informative figure (with labels) and informative text (describe what is happening in the figure!)
3. What did you learn about astro/ML/coding broadly?
4. What are your goals for tomorrow?

Classifying lensed AGN alerts

Classifying Alerts - Group 2

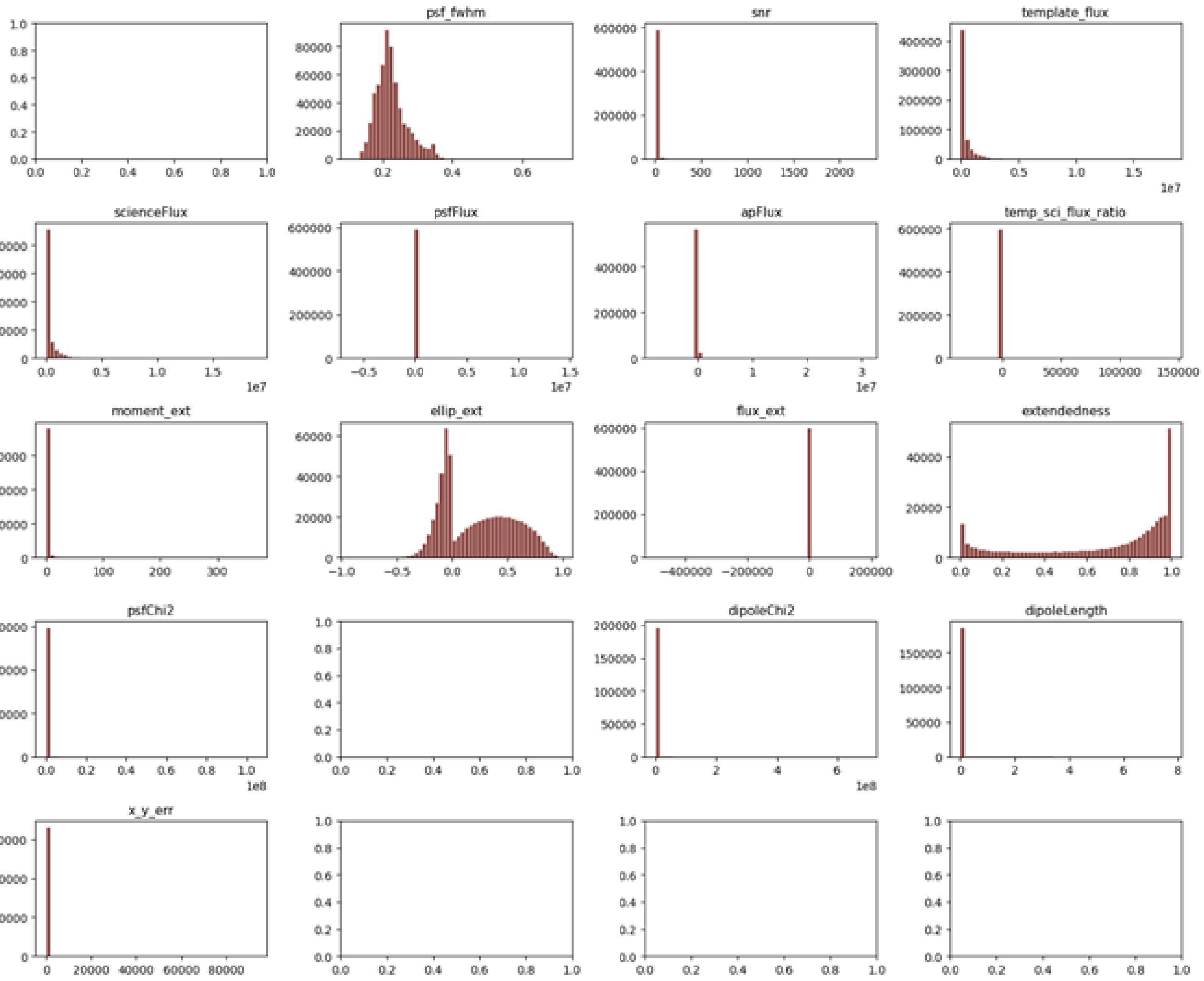
What question were you trying to answer today?

What features distinguish LAGNs from non-LAGNs?

What We Learned

- Learned how to access data with Google Colab/Drive
- The science behind AGNs and dipoles
- The basics of difference imaging analysis
- How to better troubleshoot by isolating lines of code and testing them on our own
- How to prevent data leakage by carefully choosing which columns belong in our features dataframe
- There are a lot of subjective choices that have to be made when cleaning up data

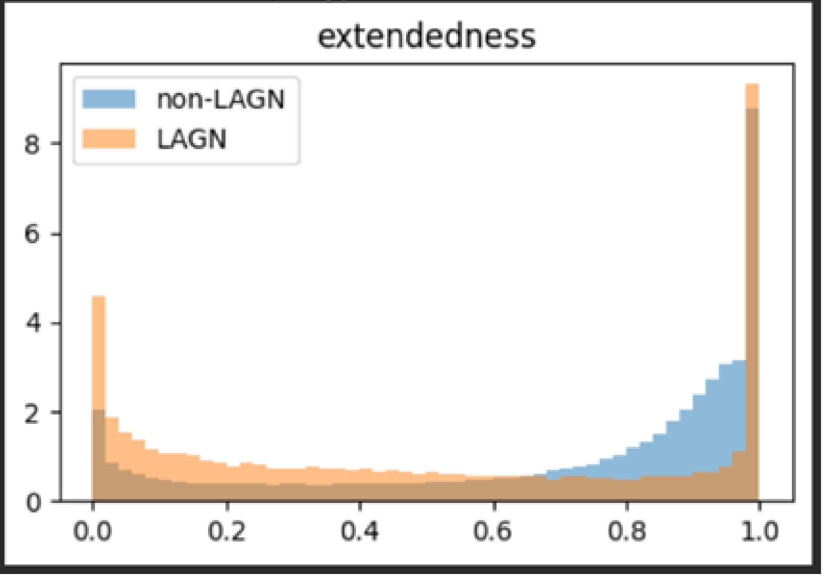
Category	Count of NaNs
psf_fwhm	405
snr	0
template_flux	0
scienceFlux	0
psfFlux	0
apFlux	153
temp_sci_flux_ratio	0
moment_ext	207587
ellip_ext	0
flux_ext	153
extendedness	307961
psfChi2	0
dipoleMeanFlux	0
dipoleLength	0
x_y_err	265469



Goals for Tomorrow

- Figuring out what to do with NaNs **(imputing, removing)** and outliers **(sigma clipping)**
- **Plotting histograms for features**, separating rows that were flagged as lensed AGN vs. not lensed AGN
- Get a working model on the cleaned data!

```
--- extendedness ---
dtype: float64
missing: 51.5%
unique values: 285165
count    2.902510e+05
mean     6.779127e-01
std      3.336850e-01
min      7.248779e-11
25%      4.257406e-01
50%      8.268421e-01
75%      9.548293e-01
max      1.000000e+00
Name: extendedness, dtype: float64
```



Finding TDE Host Galaxies

-Eliminated outliers, 5σ

-Cleaned NaN and Inf

-Filtering by redshift ($z < 0.3$)

-Seaborn pair plots

-Labeled each galaxy by QBS (Quiescent Balmer Strong) Galaxy Conditions → 5948 galaxies

- Lick $H\delta_A$ > 1.3 Å in absorption
- Hα EQW < 5 Å in emission
- Hα emission flux > 0 & <1000

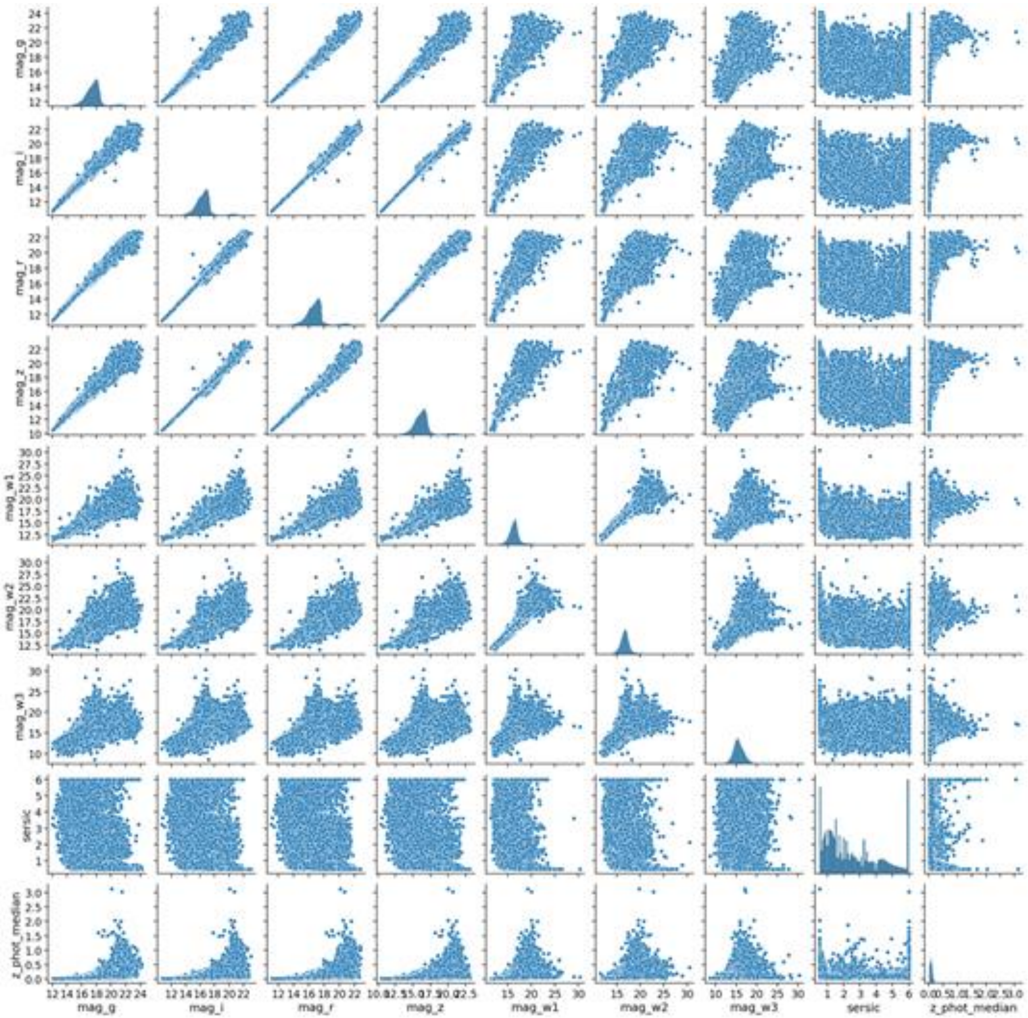
Tomorrow:

-RandomForest training and validation

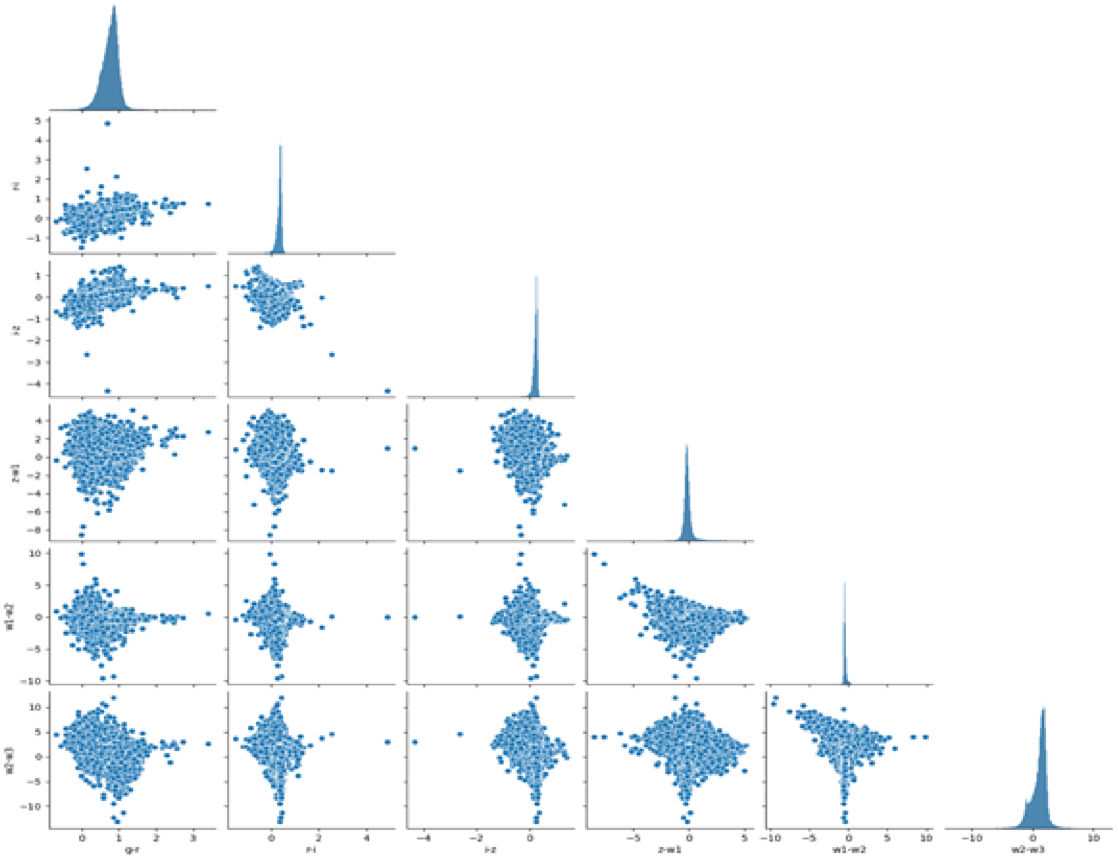
-Dataset split

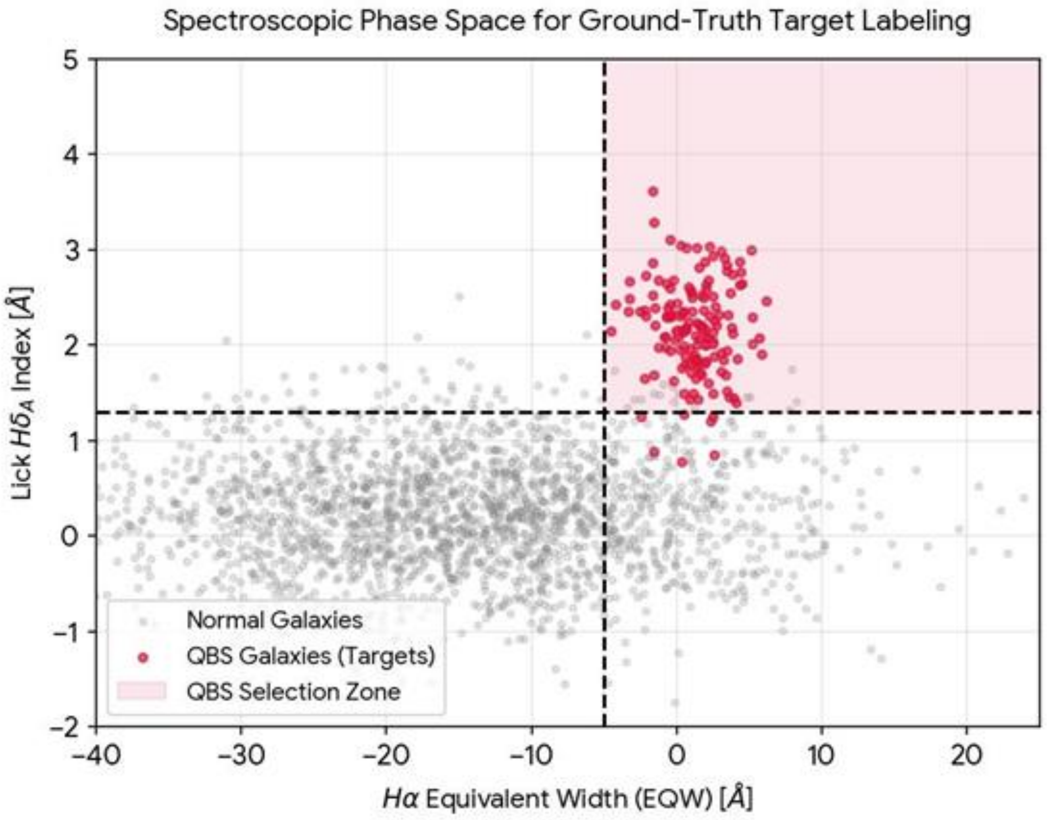
-Cross Validation

-Determine best model



- I learned how to use pre-clean data prior to training the model
- Most outliers: z-w1, w1-w2, and w2-w3 have the largest number of extreme values g-r and r-i look cleaner, with fewer extreme points compared to the others.
- My goals for tomorrow would be to understand the data itself and how its being interpreted.



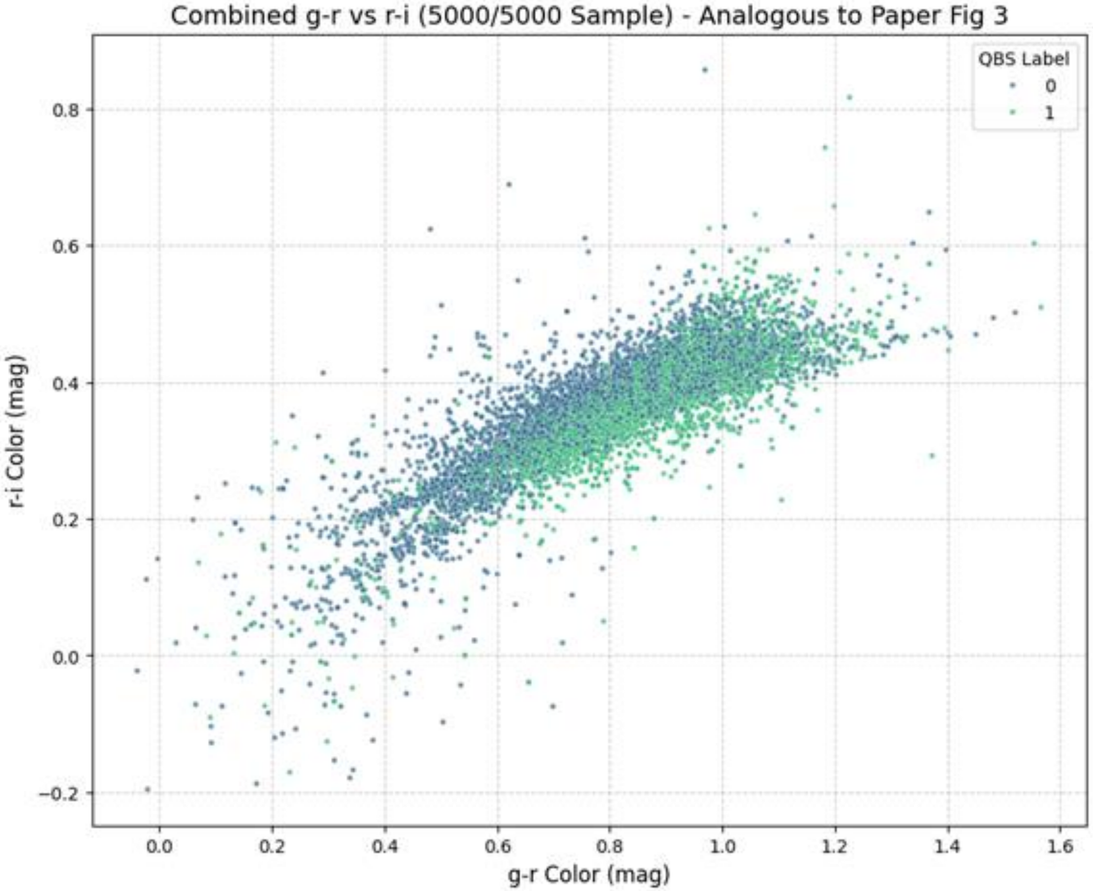


Ground-Truth Target Labeling: Identifying QBS Galaxies

Defining Our Ground Truth: The sharp spectroscopic boundaries we applied created a perfectly clean plot, giving us a definitive, reliable set of target labels (\(y = 1\)) to work with.

The Looming Data Deluge: Upcoming wide-field surveys like the Rubin Observatory won't have this detailed spectroscopic data available; they will only provide us with raw photometric colors (\(u, g, r, i, z, y\)).

Tomorrow's Challenge: Our next goal is to train a machine learning model to bridge this gap, learning to accurately flag these exact target galaxies using only that limited photometric light profile.



Photometric Confusion

The Photometric Blur: While our spectroscopic plot was clear, moving to raw colors ($g-r$ vs. $r-i$) completely blends normal and target galaxies because broad filters dilute the sharp, diagnostic line features.

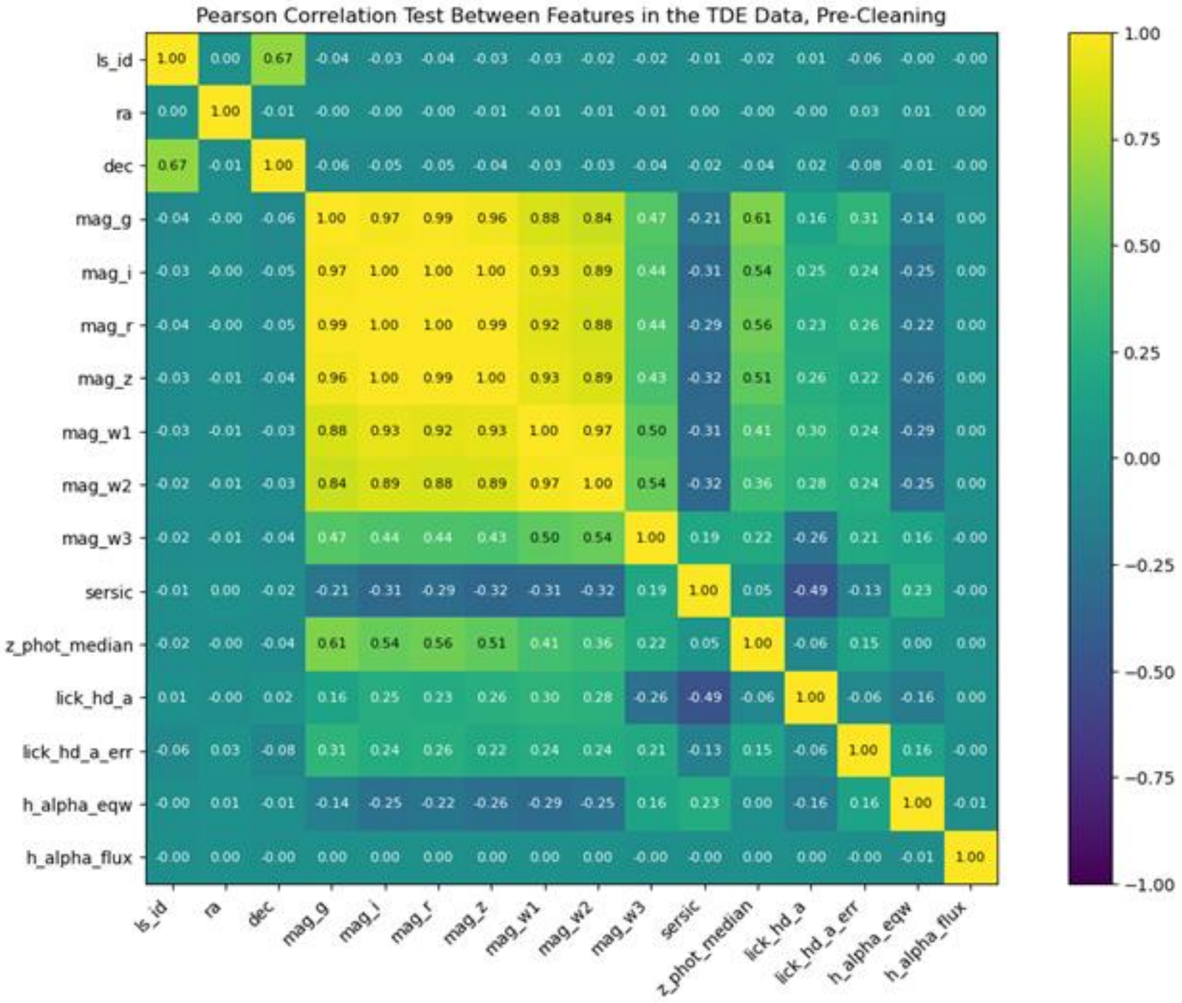
Why Human Eye Fails: Because the populations overlap so heavily in this scatter plot, it is physically impossible for a human to draw a clean, manual boundary line to separate them.

The Machine Learning Fix: This is exactly why we need machine learning; algorithms like Random Forest can dig past this 2D blur by analyzing all 6+ physical features at the exact same time.

Rei’s Data Exploration

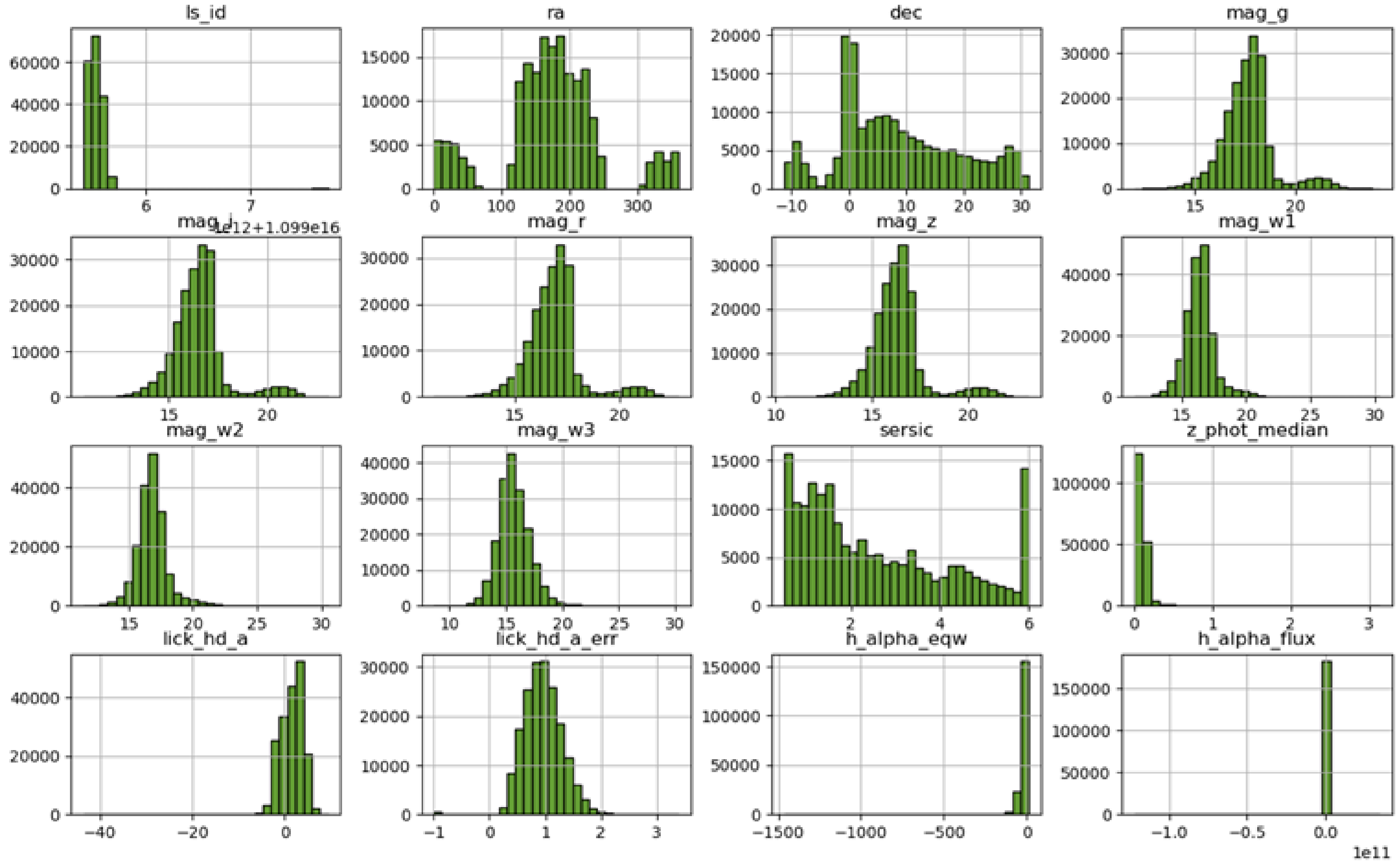
To improve our data science skills, it is important to understand the units of the columns

Column Name	Units
RA (Right Ascension), DEC (Declination)	Degrees
mag_g, mag_i, mag_r, mag_z, mag_w1, mag_w2, mag_w3	AB Magnitude
Sersic	Unitless
z_phot_median	Unitless
lick_hd_a	Angstroms
lick_hd_a_err	Angstroms
h_alpha_ew	Angstroms
h_alpha_flux	erg/s/cm^2/Angstroms

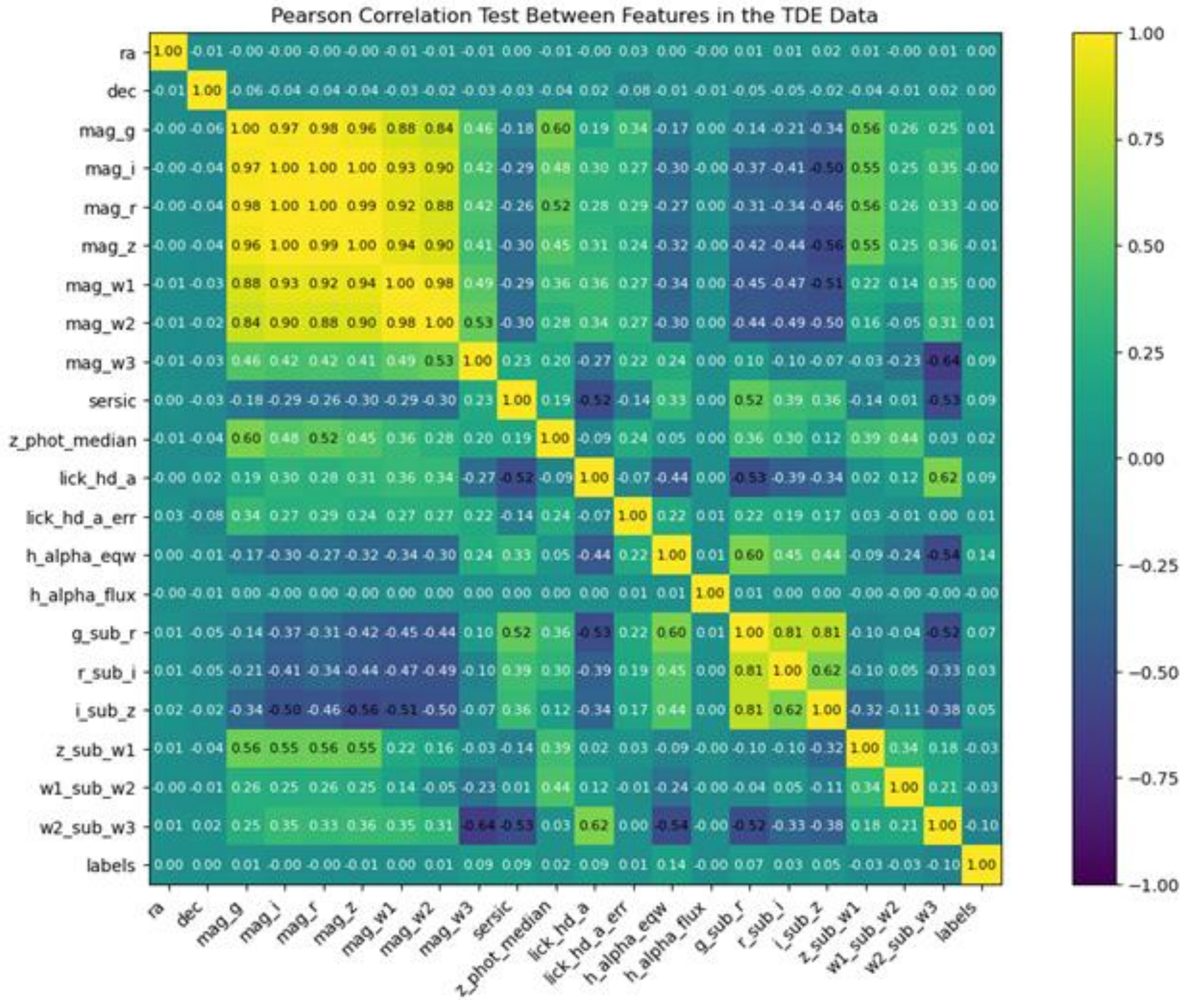


When plotting the Pearson correlation test, there are high correlations between all of the band magnitudes. This is important to keep in mind for feature engineering.

The distribution of the data, pre-cleaning, shows that most features do not have a normal distribution and even some bimodality



It's also important to take note of the different ranges! Will need to scale these features!



After filtering, cleaning, and calculating metrics, there is still a strong correlation between the band magnitudes. As in the previous slide, it's important to keep in mind the relationship of these bands for feature engineering

Goals for tomorrow:

- Create the Random Forest model, but explore many different methods:
  - What happens if I change the scoring method?
  - What happens if I adjust the size of the training/validation/test sets?
  - Comparing the model results for when the features are scaled vs. not scaled.
- Get a deeper understanding of model metrics:
  - Understand how different models affect the Precision, Recall, F1, and ROC-AUC scores

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cosmicAI Workshop Day 2

Data prep and random forests

On the plate 🍽️ for today

1. Set up the RF classifier, *ask Padma if you get stuck (I have some semblance of the solution!)*
2. Spend the first 5 minutes making a list of what you want to look into today
3. Go forward with the final cleaned data you have and come back to data prep later if needed
4. Work until break (10:30am) – come back at 11 – continue working until 12:30 (hopefully the RF classifier is set up) – post lunch is free form exploration – final presentation

Post lunch

1. 2 - 3:30pm – work on your random forest classifier results – experiment with different hyperparameters, try to beat the baseline (0.5 – random classification)!
2. 3:30 - 4pm – make your presentation – think about *what you are going to say*, there should be a coherent narrative and informative slides
3. Presentation guidelines
    a. Your presentation should introduce the science problem
    b. The data problem (why do we need ML?)
    c. The method you are going to use
    d. **Your preliminary results – any first result is a good result!**
    e. Future work
4. 4 - 4:45pm – final presentations; final survey in the last 15 minutes